Developing Tullomer in Drone Applications

Report by Sam Linnington, 10/15/2026, for use by Z Polymers

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Abstract

This report documents the development and future testing of a modular 5-inch FPV drone platform created to evaluate Z Polymers Tullomer against incumbent 3D-printable materials. The purpose of testing against incumbent materials is to measure whether Tullomer can outperform materials already at the cutting edge in any useful metric; thereby justifying reasonable investment into future research and testing for drone applications. The planned evaluation is encompassed by 3 core tenets: structural efficiency; RF performance; and vibration behavior. The core materials recommended as incumbents against which to benchmark Tullomer are Polycarbonate (PC) and Carbon Fiber reinforced Nylon (CF-PA). Acrylonitrile Styrene Acrylate (ASA) as well as Polyethylene Terephthalate Glycol (PETG) could be used as well if a more diverse range of comparisons/controls is desired.

As of this report, the platform remains in a flight-validated proof-of-concept state: the drone has been assembled, wired, and flown successfully under a functional flight-controller and radio-link configuration. This comes after several rounds of troubleshooting with semi functional old parts, and is now ready to be standardized for experimentation. Material testing has yet to be executed; this report establishes the finalized hardware baseline by resolving outstanding platform-readiness issues and defining the testing methodology future material evaluations could follow.

Ultimately this report serves its purpose twofold: first, as a status summary for strategic decision-making in future Tullomer testing; and second, as a technical guide for whomever it may concern driving this project in the future. The anticipated outcome of this advisory is to clarify the status and potential future of Tullomer in drone applications, and — pending confirmation through testing — achieve quantifiable value propositions via signal, structural, and vibration comparisons across evaluated materials.

Current State

Bill of Materials

- Leopard FPV Drone
- Signal
  - HappyModel EP2 TCXO (2.4GHz ELRS) Reciever
  - Radiomaster Pocket (EdgeTX, internal ELRS module) (2S LiOn 18650s, USB-C in-radio charging)
- Controls
  - SpeedyBee F405 Mini BLS 35A 20x20 Stack
- Power Transfer Systems
  - EMEPOVGY B6 Lipo Battery Balance Charger 80W 6A
  - 35V 470µF capacitor
  - Tattu R-Line 4S LiPo Battery (14.8 V nominal / 850 mAh / 95C)
  - MR30 male and female connectors for motor to ESC power
  - XT30 male and female connectors for battery lead
- Flight Components
  - XING2 1404 3800KV motors (old) / Surpass Hobby S2207 2550KV Motors (current)
  - 8pcs HQProp 5X4.8X3 V1S 5048 5 Inch 3-Blade Propellers Set - Poly Carbonate
- Hardware Connections
  - Mxuteuk 540PCS M2 M3 Metric Screws Assortment Set
  - Kadrick 420Pcs M3 M4 M5 Threaded Inserts, Assortment Kit

Component Review

* Convention

A series of common abbreviations to improve readability for anyone reading this:

- RX = Receiver
- TX = Transmitter
- FC = Flight Controller
- ESC = Electronic Speed Controller
- RSSI = Received Signal Strength Indicator

Geometry

The standard 5-inch propeller size balances two competing needs: small enough to be practical in constrained spaces, and large enough that the results can be generalized to common commercial and defense FPV platforms. The frame geometry is derived from an open-source design posted on Printables, the STL files of which were used for the current proof-of-concept iteration. Several editable Onshape models were developed for the arm geometry so that future designs (and by extension, the printed material used) can be revised or swapped without affecting the rest of the frame, isolating material performance as the sole varibale for testing. Developing editable models of top and bottom frame elements were not a goal during this time.



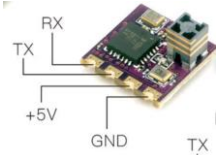


Receiver (RX) & Transceiver (TX)

Flight dynamics are controlled via radio frequency, which is handled by a HappyModel EP2 TCXO receiver (directly soldered to the FC) paired with a Radiomaster Pocket transmitter (with an internally built in radio model), communicating over ExpressLRS (ELRS) on the 2.4GHz band. Both the RX and TX must run matching firmware versions for the link to function correctly.

- RX wiring: 4V5, G, R2, T2 pads (UART2 on flight controller)
- RX mode (Betaflight): Serial, via UART
- Serial receiver provider: CRSF
- Current Firmware (RX and TX, confirmed matching): 3.3.1
- Transmitter battery: 2S 18650 Li-ion, charged in-radio via USB-C
- Transmitter ADC filter: OFF

RX:



TX:

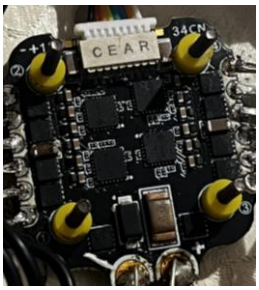


Control Stack

Flight control and motor power are handled by a SpeedyBee F405 Mini BLS 35A 20x20 stack, which combines the flight controller (FC) and electronic speed controller (ESC) into a single compact unit connected by a set of wire connectors. The ESC currently implemented is fully functional and has been the primary unit flown throughout testing. A duplicate ESC from the same stack was wired to the older XING2 motors, but that pairing showed inconsistent motor behavior (likely an issue with the older motors, their connectors, or both). The stack runs Betaflight (for anyone picking up the project: get familiar with Betaflight configurator and ELRS configurator) and communicates with the ESC using a standard digital motor protocol and other settings detailed below.

- Motor/ESC protocol: DSHOT600
- 3D Mode: OFF
- MOTOR_STOP: enabled
- Airmode "Always On": disabled
- Motor idle: 5%
- ARM (in Modes tab on Betaflight): mapped to AUX1 (aka SA on controller), range 1700–2100

FC: 

ESC: 

Motor & Power Systems

The current build uses Surpass Hobby S2207 2550KV motors, paired with HQProp 5x4.8x3 propellers and powered by 4S LiPo battery packs. Power runs from the battery through the control stack to each motor, with a capacitor (35V 470µF) placed at the power input to help smooth voltage delivery to the electronics. Battery connections use XT30 connectors, and motor-to-ESC connections use MR30 connectors (more on this in Recommendations & Troubleshooting). An earlier version of the platform used iFlight XING2 1404 3800KV motors, but later sections will detail why this is advised against for future development.

Troubleshooting & Recommendations

Retired Hardware

Several components from earlier in the project have since been replaced and are no longer part of the current build. First, the original iFlight XING2 1404 motors were retired after repeated inconsistency, potential burnout, and a desire for a larger more generalizable platform. The overall recommendation is to cease developing a smaller motor/frame class alongside the current build. A prior flight controller was retired after a broken solder pad interrupted the receiver connection, and a damaged multi-wire connector between an earlier ESC and flight controller was bypassed by soldering the ground wire directly to the board. Early troubleshooting on the link system pointed to the previous Radiomaster Pocket as a root cause of failure, since the TX-side ELRS Lua script (the tool meant to set up and control binding) repeatedly froze and refused to connect over USBC or Wi-Fi for updates. Additionally, link instability (between RX & TX) had been observed several times across multiple different RXs and FCs, ruling out those components as the primary fault. To combat this a new TX unit was purchased. After obtaining a new TX unit it was later found out that the Radiomaster pocket has two USBC ports, the lower for power and the upper for data. My recommendation would be to attempt firmware reflashing and updates over a USBC connection on the old TX using ExpressLRS Configurator in order to salvaging its usefulness. These items are noted here for historical context and are not present in the configuration of the current platform.

Outstanding Issues

The unresolved issues with the current platform fall into two overarching categories: RX & TX auto-linking and FC BLE connectivity.

- First, the receiver requires manual rebinding to the transmitter after every power cycle, rather than reconnecting automatically as it should. Its worth noting that the receiver currently in place is the second of three, and the described issue could potentially be resolved by simply soldering in the fresh RX board.
- Second, the FC was never able to connect to the SpeedyBee companion app over Bluetooth. This has been tested and reproduced across multiple flight controller units, suggesting either a recurring hardware limitation or an incompatibility specific to the phone used for testing, rather than a one-off defective unit.

Future Upgrades

Bare Minimum

- Apply a 75% motor output limit in Betaflight's PID tuning tab, to avoid over-stressing the current connectors and ESC until further upgrades are implemented.
- Swap in the new receiver already on hand, which may resolve the rebinding issue described above.
- Replace MR30 motor connectors with MR60 connectors and solder them between the motors and ESC to modularize the setup (necessary if drone arms are going to be swapped frequently). The current MR30s possessed (originally purchased to be used with XING2s) are rated for 15A continuous, well under the 47A the S2207 motors can draw; a mismatch that risks overheating and connector failure.

Highly Recommended

- Switch from HQProp 5x4.8x3 to HQProp 5x4x3 propellers. The lower pitch bites less air, reducing current draw, heat, and battery wear. Spare propeller nuts and washers should be purchased alongside this change, as they are easy to lose and inexpensive to replace.
- Switch from XT30 to XT60 connectors on both the battery packs and ESC lead. XT30 is rated for 30A continuous, while the current platform can draw well over 100A under load. XT60 connectors raise the ceiling to 60A, and should be paired with a switch to 4S 1300mAh battery packs; the current 850mAh packs are electrically compatible but undersized for sustained flight on this motor/propeller combination, risking voltage sag and accelerated wear.

Long-term Sustainability

In the event that this project makes real headway towards commercial, industrial, or defense applications, upgrading the control stack (to 30x30) to preserve long-term function for testing is recommended. At full throttle, the current 20x20 stack's ESC (rated 35A continuous, 45A for 5 seconds) is estimated to draw roughly 46A across all four channels at once. well beyond its rated capacity, risking overheating, desyncs, or a mid-flight failure during sustained use. Additional considerations could be made depending on future directions. These include but are not limited to: integrating a video and GPS telemetry system; adding a payload for higher fidelity comparison to military and industrial applications; utilizing a larger frame for broader generalizability (7 inch more standard for longer distance sustained flight); leveraging multi material printing (e.g. Tullomer encased in TPU, etc) for potentially improved structural stability; and testing in a large range of environments to further explore material versatility (e.g. extreme high and low temperatures, as well as high moisture and amphibious applications).

Material Testing

Radio Frequency Performance

RF testing is intended to determine whether Tullomer offers a measurable advantage over incumbent materials in radio signal transmission and reception, with particular emphasis on comparison against carbon-fiber-reinforced nylon (CF-PA). Carbon fiber is electrically conductive, and is commonly understood in UAV communities to interfere with RF signals when placed near RX & TX components or in a signal's path. While this often requires pilots to work around CF with creative placement or shielding, Tullomer's potential RF transparency relative to CF represents the most direct and demonstrable value proposition this project is positioned to prove. The underlying material property (dielectric behavior vs. conductivity) is something that can be shown clearly, even without full drone-level testing.

To start, the quickest avenue to measurable benefit would be bench-level testing that isolates the material property directly. There are several options on testing implementation, which vary based on the desired failure mode tested and the room for investment into specialized signal analysis equipment. It should also be noted that if a video system were to be implemented, that would utilize of a different frequency (5.8 GHz) than standard ELRS, which would grant another data point on which to test Tullomer.

Option 1 is the only of the 3 offered in this report that utilizes strictly the hardware already used in the project. To execute, a tester would use the existing ELRS RX & TX, separated by a shield or enclosure of the chosen materials (absolutely uniform, in all dimensions and print parameters, material being the only independent variable), to measure Received Signal Strength (RSSI), Link Quality (LQ) (including packet loss) and maximum reliable control distance. The unfortunate truth of these test methods is that they are system-level, protocol-processed outputs (subject to frequency hopping, error correction, and automatic gain control). This makes said test the most prone to setup error and the least precisely quantitative (and therefore the hardest to claim credible in a professional standing), though still valuable; fast, free, and easy to demonstrate.

Option 2 requires purchase of more specialized analysis tools: a NanoVNA capable of operating at 2.4GHz, along with two matched test antennas. To execute, a tester would place the sample directly between the NanoVNA's transmit and receive paths and measure how much signal is lost passing through it (expressed as insertion loss in decibels). Because the NanoVNA generates its own calibrated test signal and reads the result directly (with no receiver firmware, frequency hopping, or automatic gain control sitting in between) this test isolates the material's physical RF-blocking property with a precision and repeatability the existing ELRS hardware cannot match. This makes it the

most direct, credible, and industry-defensible way to state a specific quotable difference between materials, at the cost of a small one-time equipment investment and a setup the team hasn't previously used.

Option 3 uses the same NanoVNA as Option 2, but in a different measurement mode; rather than transmitting through a material, it measures voltage standing wave ratio (VSWR) (AKA how much signal reflects back from a fixed antenna when a material sample is placed at a set distance and orientation nearby). To execute, a tester would mount a reference antenna, then introduces each material sample at a repeatable position near it, recording any shift in the antenna's reflected signal. This measures whether a material's proximity distorts or detunes the antenna itself, a set up much closer to how frame material actually sits relative to an antenna on a finished drone. Since it uses the same instrument as Option 2, it requires no additional equipment investment once said NanoVNA is acquired and it produces the same caliber of precise, repeatable, instrument-grade data, making it a natural companion test to Option 2 rather than a separate cost.

Antenna recommendation: 'HappyModel 2.4GHz Moxon SMA anten**na**' which is tuned for the same 2.4GHz ELRS band and is the same antenna family actually used in FPV ELRS hardware, so bench measurements are high fidelity (You'd want two of these (~$10–15 total), one per NanoVNA port, so the same antenna type sits on both the transmit and receive side for Option 2, and a single one for the reflection-only Option 3).

NanoVNA recommendation: **a** 'LiteVNA-64' ($150-250) or 'NanoVNA-F V3' ($200-300) would be best suited, as they both cover up 1-6Ghz, the range needed to evaluate both the 2.4 Ghz ELRS band and the 5.8 Ghz VTX band (enabling future research development).

NOTE: Should a VTX system be added to the test platform, (I'd recommend looking into analog or HDZero ecosystems) RF evaluation should be extended to the 5.8GHz video band using the same Option 2/3 methodology; this would require matched 5.8GHz test antennas (such as 'RunCam 5.8GHz antenna' (~$20-30) a comparable antenna to the original Leopard FPV build), in addition to the 2.4GHz equipment specified above.

Structural Efficiency

Structural testing is intended to determine how Tullomer compares to incumbent printable materials in real-world impact resistance; specifically, how much structural performance each material delivers relative to its weight and cost. Since frame arms are both the most failure-prone component in a crash and the primary variable this project is designed to isolate, the outlined tests should directly address whether Tullomer offers a meaningful durability advantage.

Test Metrics

- ARM manufacturing parameters (standardize across all materials): printer, nozzle, layer height, wall count, infill, drying procedure, print orientation, and annealing/post-processing
- ARM mechanical parameters (will necessarily vary): material, wall thickness, infill, fiber orientation, geometry/print temperature
- Resulting mass and chosen drop heights
- Material ($/g) and part (total) cost
- Deflection under applied load

Each material's arm geometry should be evaluated using a vertical drop rig, (by hand or a mechanism) at matched heights across a fixed set of surfaces (rubber mat, plywood, concrete) and a fixed set of impact orientations (belly, arm-tip, front-corner) to simulate a range of real-world impact conditions and capture how each material responds to different realistic crash angles. Since identical print geometries do not guarantee identical mass across materials, dropping each material from the same height does not deliver the same impact energy, and therefore mass must be measured and logged for every part so that actual impact energy (mass × 9.81 × drop height) can be calculated per trial during analysis. Results should be reported as deformation relative to impact energy delivered, not deformation relative to drop height alone, to avoid misattributing a mass-driven energy difference to material performance.

Each drop should record a slow-motion video from side-view, with a ruler or grid placed in frame to allow direct measurement of deformation. After each drop, post-impact inspection photos should be taken and logged against a consistent observation checklist (e.g. a cracked arm, loosened module, damaged fastener insert, permanent bend, an electronics bay intrusion, etc).

Two additional methods are worth considering as extensions once baseline drop testing is established, rather than required for an initial pass

1. using the flight controller's onboard accelerometer and blackbox log during crash events to capture how much shock actually reaches the electronics bay (connecting structural performance directly to the Vibration Behavior testing core),
2. Direct tensile testing of individual arm geometries in isolation, which would provide a cleaner, more controlled strength measurement than impact testing alone, decoupled from drop-rig variability like bounce, surface inconsistency, or imprecise release height.

Vibration Behavior

Vibration testing is intended to determine whether frame material measurably affects how vibration and resonance propagate through the drone during flight; specifically, whether Tullomer dampens, transmits, or amplifies vibration differently than incumbent materials. Significance lies in that excessive frame-transmitted vibration degrades flight performance (noisy gyro data, filter strain) and accelerates wear on electronics and fasteners over time, so a material's vibration behavior heavily affects a vehicles long-term viability.

This test should rely exclusively on the flight controller's existing onboard accelerometer and gyroscope, logged via Betaflight's built-in Blackbox feature during controlled flights, as this capability is already present on the current stack and is the standard, industry-accepted method for this variety of analysis). Each material's arm geometry should be flown through the same controlled flight pattern (consistent throttle ramps, maneuvers, and duration) so that logs are directly comparable across materials, with flight conditions (battery charge, wind, pilot input) held as consistent as practically possible between trials. Resulting logs should be analyzed using Betaflight Blackbox Explorer, Betaflight's standard first-party analysis tool.

Metrics

- Peak acceleration amplitude
- RMS (root-mean-square) acceleration
- Dominant vibration frequencies
- Frequency spectrum (FFT/spectral analysis)
- Spectrogram vs. throttle/RPM
- Identification of persistent resonance frequencies

The anticipated result is a comparative vibration profile across materials, identifying whether Tullomer exhibits meaningfully different damping or resonance characteristics than incumbent materials and if it offers any vibration-related durability or performance advantage in addition to any structural or RF findings.

Supplemental Media

Video:

Matrix Line Drone - YouTube; Military target equivalent

- 7:00 Close look
- 18:50 Flight Demo
- First half of video definitely worth watching for overall context

Optional Reading Material:

- Carbon fibre for applications in aerospace: A review
- Comparison of Materials for Unmanned Aerial Vehicle
- Vibration Characteristic Analysis and Dynamic Reliability Modeling of Multi-Rotor UAVs
- Effects of FPV Drone Frame Materials on Thermal Conditions of Motors Under Extreme Payloads: Experimental and Numerical Analysis
- An investigation on the structural vibrations of multi-rotor passenger drones

Additionally useful:

- I would personally recommend a few general intro videos on FPV, Onshape, and Betaflight Configurator if previous experience is inadequate/nonexistent
- Utilize experts in the field (e.g. Joshua Bardwell - YouTube / JoshuAI Bardwell — FPV Drone Assistant)

Implementation Strategy

1. Finalize platform readiness. Evaluate and apply upgrades as tiered in Troubleshooting & Recommendations section in priority order. Testing should not begin until at least the bare-minimum tier is complete, given the connector/ESC overcurrent risks already identified.
2. Resolve receiver rebinding issues using the new receiver already on hand, and confirm flight-level control benchmarks are stable before relying on this platform for RF testing.
3. Decide explicitly which materials will be tested (refer to Abstract section), and whether an attempt at salvaging the old TX is worth while (refer to Retired Hardware section)
4. Evaluate options for RF testing, and accordingly obtain RF bench testing equipment if desired (heavily recommended).
5. Standardize fabrication and mechanical parameters across all geometric elements (print parameters, screw types, fastener placement, and assembly process) in consultation with Tullomer's process expert (Nezy).
6. Execute RF testing as selected
7. Execute structural testing
8. Execute vibration testing
9. Compile comparison and recommendation